

CSA0501
DATABASE MANAGEMENT SYSTEMS
LAB EXPERIMENT

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EXP 7:

AIM

To view the records from the tables using SELECT commands with GROUP BY, HAVING, and ORDER BY clauses.

ALGORITHM

1. Start.
2. Create a database and select it.
3. Create STUDENT, FACULTY, and StudentScores tables.
4. Insert sample records into each table.
5. Use GROUP BY with aggregate functions to summarize data.
6. Use ORDER BY to sort records in required order.
7. Display and verify the output of each query.
8. Stop.

COMMANDS

sql

```
CREATE DATABASE dbms_lab7;
```

```
USE dbms_lab7;
```

```
CREATE TABLE STUDENT (...);
```

```
CREATE TABLE FACULTY (...);
```

```
CREATE TABLE StudentScores (...);
```

```
INSERT INTO STUDENT VALUES (...);
```

```
INSERT INTO FACULTY VALUES (...);
```

```
INSERT INTO StudentScores VALUES (...);
```

```
SELECT CourseDesc, COUNT(StudentNo) FROM StudentScores GROUP BY CourseDesc;
```

```
SELECT StudentNo, COUNT(CourseDesc) FROM StudentScores GROUP BY StudentNo;
```

```
SELECT Name, Gender, MobileNo FROM STUDENT ORDER BY RegNo ASC;
```

```
SELECT * FROM FACULTY ORDER BY DOJ ASC;
```

QUESTIONS

GROUP BY – HAVING

1. How many students are registered for each course? Display the course description and the number of students registered in each course.
2. How many courses did each student register for?

ORDER BY

1. Retrieve Name, Gender, Mobile No of all the students in ascending order of Reg No.
2. List the faculty members in the order of older faculty first.

OUTPUTS:

```
mysql> INSERT INTO StudentScores VALUES ('S003', 'Networks');  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT CourseDesc, COUNT(StudentNo)  
-> FROM StudentScores  
-> GROUP BY CourseDesc;
```

CourseDesc	COUNT(StudentNo)
DBMS	2
OS	2
Networks	2

```
3 rows in set (0.05 sec)
```

```
mysql> SELECT StudentNo, COUNT(CourseDesc)
-> FROM StudentScores
-> GROUP BY StudentNo;
```

StudentNo	COUNT(CourseDesc)
S001	2
S002	2
S003	2

3 rows in set (0.00 sec)

```
mysql> SELECT Name, Gender, MobileNo
-> FROM STUDENT
-> ORDER BY RegNo ASC;
```

Name	Gender	MobileNo
Kavya	F	9988123456
Ravi	M	9876501234
Meena	F	9012345678

3 rows in set (0.02 sec)

```
mysql> SELECT *
-> FROM FACULTY
-> ORDER BY DOJ ASC;
```

FacNo	FacName	Gender	DOB	DOJ	MobileNo	DeptNo
F001	Ganesh	M	1978-04-11	2005-07-01	9988001122	CSE
F002	Lakshmi	F	1982-09-23	2009-01-15	9876554433	ECE
F003	Anand	M	1988-11-05	2013-03-20	9012334455	IT

```
3 rows in set (0.01 sec)
```

RESULT:

The records from the tables are displayed using SELECT commands with GROUP BY, HAVING and ORDER BY.